

5 Extractors (Window t)

1. Joint AAGCN $\rightarrow \mathbf{p}_1^{(t)}$

2. Bone AAGCN $\rightarrow \mathbf{p}_2^{(t)}$

3. Joint-Motion AAGCN $\rightarrow \mathbf{p}_3^{(t)}$

4. Bone-Motion AAGCN $\rightarrow \mathbf{p}_4^{(t)}$

5. Transformer Mix $\rightarrow \mathbf{p}_5^{(t)}$

$\{\mathbf{p}_k^{(t)}\}_{k=1}^5$

Stage 1: Window SLSQP

$$\mathbf{P}^{(t)} = \sum_{k=1}^5 w_k^* \mathbf{p}_k^{(t)}$$

$$\min_{\mathbf{w}} \mathcal{L}_{\text{val}} \text{ s.t. } \sum w_k = 1, w_k \geq 0$$

$\mathbf{P}^{(t)}$

Stage 2: Video Consensus

Aggregation across W Windows:

$$\bar{\mathbf{P}} = \frac{1}{W} \sum_{t=1}^W \mathbf{P}^{(t)}$$

$\bar{\mathbf{P}}$

Prediction $\hat{y}$

$$\arg \max_c \bar{\mathbf{P}}_c$$

Video-Level